Arista Access Node MAR-ACC-36 QoS Congestion Event Root Cause Analysis and Resolution SOP

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7084

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve Critical QoS congestion events on the Arista Access node MAR-ACC-36 at the Marathahalli Core site. The failure domain under investigation is rooted in the occurrence of microbursts, interface tail drops, and QoS queue starvation. This phenomenon is primarily attributed to the interplay of QoS policies and the inherent characteristics of the Access node's hardware and software architecture. Specifically, the Arista EOS (Extensible Operating System) manages the node's resources, including its 40GE interfaces, to ensure efficient packet forwarding. However, when QoS policies are misconfigured or improperly set, they can lead to queue starvation, resulting in the tail drop of packets and subsequent congestion events. Furthermore, the occurrence of microbursts can exacerbate this issue by injecting high-bandwidth, low-latency traffic into the network, thereby overwhelming the node's processing capabilities. Therefore, this SOP aims to identify and rectify the root cause of the congestion event by analyzing the QoS policies and interface configurations, and implementing corrective measures to prevent similar occurrences in the future.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and that all necessary personnel, including remote hands, are dispatched to the location.

- Check the status of the redundant link to ensure that it is operational and functioning as expected.

- Confirm that all necessary network equipment, including the Arista Access node MAR-ACC-36, is powered on and responding as expected.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to queue starvation	High
Interface configuration issues resulting in tail drops	Medium
Hardware-related issues affecting the node's processing capabilities	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: Policy name, Class name, Queue name, Rate, Burst, Priority, Description: This command displays a comprehensive list of all QoS policies defined on the node, allowing us to identify any potential misconfigurations or policy conflicts.
```

```
Command: show interface 40GE 1/1, Expected Output: Interface name, Description, MTU, Speed, Duplex, Flow-control, Tx/rx packets, Tx/rx bytes, Description: This command provides detailed information about the 40GE interface, including its configuration settings and operational statistics, which can help us pinpoint interface configuration issues.

Command: show queue statistics, Expected Output: Queue name, Packets, Bytes, Dropped packets, Dropped bytes, Description: This command displays queue statistics, enabling us to identify any queue starvation issues and assess the impact of QoS policies on interface performance.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy all, Description: Review the output of the show qos policy all command to identify any misconfigured policies or policy conflicts. If any issues are found, update the policies to ensure they are correctly configured.

Step 2: Step: Configure interface to optimize performance, Command: interface 40GE 1/1, Description: Modify the configuration of the 40GE interface to optimize its performance, taking into account the results of the previous diagnostic commands. This may involve adjusting the MTU, speed, duplex, and flow-control settings to ensure optimal interface operation.

Step 3: Step: Update QoS policies to prevent queue starvation, Command: set qos policy ..., Description: Update the QoS policies to prevent queue starvation by adjusting the rate, burst, and priority settings as necessary. This will help ensure that packets are properly handled and queuing is optimized.

Post-Check Verification

```
Command: show qos policy all, Expected Output: Policy name, Class name, Queue name, Rate, Burst, Priority, Description: Verify that the QoS policies have been correctly updated by running the show qos policy all command and confirming that the output reflects the changes made.

Command: show interface 40GE 1/1, Expected Output: Interface name, Description, MTU, Speed, Duplex, Flow-control, Tx/rx packets, Tx/rx bytes, Description: Verify that the interface configuration has been optimized by running the show interface 40GE 1/1 command and confirming that the output reflects the changes made.

Command: show queue statistics, Expected Output: Queue name, Packets, Bytes, Dropped packets, Dropped bytes, Description: Verify that the queue starvation issue has been resolved by running the show queue statistics command and confirming that the output reflects the improvements made.
```

Roll-back Procedures

- Step: Restore QoS policy configuration to previous state, Command: rollback qos policy all, Description: If the changes made during the resolution process do not resolve the issue, restore the QoS policy configuration to its previous state to revert the changes made.

- Step: Revert interface configuration to previous state, Command: rollback interface 40GE 1/1, Description: If the changes made during the resolution process do not resolve the issue, revert the interface configuration to its previous state to restore the original settings.